

Ntung Ngela Landon

Cybersecurity Researcher · AI/ML Security · Trustworthy AI

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SUMMARY AND INTERESTS

Cybersecurity researcher at the intersection of AI and security, with research spanning AI/ML security, biometric security, machine learning robustness, fairness, and trustworthy AI. My work has progressed from fairness-aware face Presentation Attack Detection (PAD) to architectural bias and the robustness of fairness evaluation under distribution shift. I also have hands-on experience in ethical hacking, penetration testing, vulnerability assessment, and malware analysis, including automated cross-platform malware analysis. Research interests include AI/ML Security; Adversarial Machine Learning; Security and Robustness of ML Models; Trustworthy AI; Computer Vision Security; Biometric Security; Machine Learning Evaluation; Malware Analysis and Detection; and Security of AI-Enabled Systems.

EDUCATION

Carnegie Mellon University

Kigali, Rwanda

MSc in Information Technology -- Cybersecurity in AI/ML

Aug 2024 -- Dec 2025

University of Buea

Buea, Cameroon

BEng in Computer Engineering -- Computer Network Engineering

Oct 2019 -- Jun 2023

RESEARCH EXPERIENCE

Graduate Research Associate

Kigali, Rwanda

Carnegie Mellon University

Mar 2026 -- Present

- Lead research on the robustness and reproducibility of fairness in face Presentation Attack Detection (PAD) under cross-dataset distribution shift.
- Design and conduct cross-dataset experiments, five-seed replication, and controlled ablation studies across CASIA-SURF CeFA, OULU-NPU, and SiW-M to evaluate the stability of fairness and model performance.
- Demonstrate that aggregate model discrimination remains relatively stable under distribution shift, while demographic fairness gaps are large and highly sensitive to experimental conditions and random seeds.
- Currently working on a Microsoft-funded cybersecurity research project involving malware analysis, controlled malware experimentation, and evaluation of malware detection robustness.
- Manuscript, *Fairness Under Distribution Shift in Face Presentation Attack Detection: A Cross-Dataset and Ablation-Based Analysis*, currently under review at *IEEE Access Journal*.

Graduate Research Assistant

Kigali, Rwanda

Carnegie Mellon University

Aug 2025 -- Dec 2025

- Led a comparative investigation of CNN and Vision Transformer architectures to examine architectural bias and demographic disparities in face PAD.
- Designed and evaluated Multimodal ViT-Tiny, ResNet18, and pretrained DeiT-S models on the CASIA-SURF CeFA dataset across African, East Asian, and zero-shot Central Asian groups.
- Demonstrated that pretrained DeiT-S achieved 97.27% accuracy and 0.86% EER, reducing the African-East Asian ACER gap to 0.13% and improving zero-shot generalization to Central Asian subjects.
- Led manuscript development and research interpretation resulting in *Architectural Bias in Face Presentation Attack Detection: A Comparative Study of Vision Transformers and Convolutional Neural Networks*, presented at the 10th International Conference on Cryptography, Security and Privacy (CSP 2026), Sapporo, Japan.

Graduate Research Intern

Kigali, Rwanda

Carnegie Mellon University

Jun 2025 -- Aug 2025

- Initiated research on demographic fairness in face Presentation Attack Detection (PAD), with emphasis on underrepresented African populations.
- Developed a lightweight PAD pipeline combining Local Binary Patterns, ethnicity-aware preprocessing, and group-specific threshold optimization using Equal Error Rate minimization.
- Achieved 95.12
- Led manuscript development resulting in the peer-reviewed publication, *Fairness-Aware Face Presentation Attack Detection Using Local Binary Patterns: Bridging Skin Tone Bias in Biometric Systems*, in the *Journal of Cybersecurity and Privacy*.

PROFESSIONAL EXPERIENCE

AI, Data Management, and Cybersecurity Intern

Geneva, Switzerland (Remote)

International Telecommunication Union (ITU)

Jul 2025 -- Nov 2025

- Conducted hands-on security assessment of an organizational platform using an existing penetration-testing report as a baseline for further vulnerability investigation.
- Explored the platform to identify additional potential vulnerabilities and security weaknesses.
- Investigated the Digital Object Architecture (DOA) and Advanced Information Management System (AIMS).
- Examined the relationship between artificial intelligence, data management, and data security in AI-enabled information systems.

Graduate Teaching Assistant -- CMU Africa Bridge Program

Kigali, Rwanda

Carnegie Mellon University Africa

Jan 2025 -- Apr 2025

- Supported undergraduate students preparing for graduate-level study in computing and technology through the CMU Africa Bridge Program.
- Assisted students with technical learning activities, problem solving, and foundational computing concepts required for transition into graduate education.
- Supported instructional delivery, student engagement, academic activities, and student learning.

Networks and Telecommunications Intern

Douala, Cameroon

Réseaux et Télécommunications (Resotel)

Sep 2022 -- Feb 2023

- Supported installation and initial configuration of surveillance camera systems to strengthen organizational security infrastructure.
- Maintained optical fiber connections to ensure reliable and secure data transmission.
- Deployed a Synology Network Attached Storage (NAS) server to improve secure data storage and accessibility.

PUBLICATIONS

Peer-Reviewed Journal Publication

Jema David Ndibwile, **Ntung Ngela Landon**, and Floride Tuyisenge. "Fairness-Aware Face Presentation Attack Detection Using Local Binary Patterns: Bridging Skin Tone Bias in Biometric Systems." *Journal of Cybersecurity and Privacy*, 6(1), 2026. [link to view](#)

Conference Paper / Preprint

Ntung Ngela Landon, Floride Tuyisenge, and Jema David Ndibwile. "Architectural Bias in Face Presentation Attack Detection: A Comparative Study of Vision Transformers and Convolutional Neural Networks." arXiv:2606.18510, 2026. [link to view](#)

Manuscripts Under Review

Ntung Ngela Landon, Floride Tuyisenge, Remy Dukundane, Emmanuel Iduh, and Jema David Ndibwile. ``Fairness Under Distribution Shift in Face Presentation Attack Detection: A Cross-Dataset and Ablation-Based Analysis." Under review, *IEEE Access*.

Jema David Ndibwile, **Ntung Ngela Landon**, and Lunodzo Mwinuka. ``Adversarial Artificial Intelligence Threats in Smart Energy Grids: When Learning Systems Learn from the Attacker." Under review, *Energy Reports*.

PROJECTS

Cross-Platform Automated Malware Analysis Pipeline

MSc Final Practicum, Carnegie Mellon University Sep 2025 -- Dec 2025

CAPE Sandbox Python Virtual Machines SIEM Malware Analysis

- Designed and implemented an automated malware-analysis pipeline for Windows and Linux samples using CAPE Sandbox and dedicated Windows and Ubuntu virtual machines.
- Developed automated sample classification and routing, directing malware to the appropriate analysis environment based on its target operating system.
- Integrated a honeypot network, event-hash processing, ingestion scripts, and CAPE Sandbox to automate malware collection, execution, and behavioral analysis.
- Generated structured analysis reports and integrated the pipeline with the CMU-Upanzi SOC/SIEM for downstream security monitoring and analysis.

Ethical Hacking and Penetration Testing

Jan 2026 -- Apr 2026

Kali Linux Nmap Burp Suite Metasploit Wireshark

- Conducted hands-on penetration testing across virtualized network environments, including vulnerability discovery, network and Wi-Fi security assessment, traffic analysis, and automated security scanning.

IoT-Based Fetal Health Monitoring System

Mar 2023 -- Jul 2023

BEng Final-Year Project

IoT Signal Processing Arduino Embedded Systems

- Designed and implemented an IoT-based fetal health monitoring system incorporating signal processing, wireless data transmission, and real-time visualization of fetal health indicators.

SKILLS

Programming	&	Python; C/C++; JavaScript; NumPy; Pandas; scikit-learn; TensorFlow;
Data		OpenCV; Matplotlib.
AI/ML		Computer Vision; Deep Learning; CNNs; Vision Transformers; Feature Ex-
		traction; Model Evaluation.
Cybersecurity		Ethical Hacking; Malware Analysis; Network Security; Biometric Security;
		Presentation Attack Detection.
Security Tools		Kali Linux; Wireshark; Burp Suite; Metasploit; Nmap; Nikto; Hydra; CAPE
		Sandbox.
Systems		Linux; Windows; Virtualization; NAS; SIEM; Optical Fiber Networks.
Development		Git; GitHub; Arduino IDE.
Languages		English (Fluent/Native); French (Intermediate)

AWARDS AND HONORS

Future Africa Leaders Award, 2024. Leadership Initiative Award, 2020.

LEADERSHIP AND SERVICE

- Ambassador, Partnership and Sponsorship Lead, Africa Forge Challenge 2.0, Future Africa Leaders Foundation.